

Chapter 1

Important Formulae

$$\lambda = \frac{h}{p}$$

$$p = mv$$

$$\hbar = \frac{h}{2\pi}$$

$$k = \frac{2\pi}{\lambda}$$

$$E = hf$$

$$f = \frac{\phi}{h}$$

$$E = \frac{k^2 \hbar^2}{8\pi^2 m}$$

$$E = \frac{n^2 \hbar^2}{8mL^2}$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

$$F(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

λ is wavelength

p is momentum of a particle

h is Planck's constant $h = 6.624 \times 10^{-34} Js$

m is the mass of a particle

moving at a velocity v

1. Find the temperature at which the probability of occupation of the energy state 0.75eV above the Fermi level is 30 percentage.

Solution:

Given,

Energy above the fermi energy level $(E - E_F) = 0.75eV$

Probability of occupation $(F(E)) = 0.3$

$$F(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

2. Find the temperature at which the probability of finding the electron at the energy state 0.25eV above Fermi level will be 20%.

Solution:

$$F(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

3. Calculate the second and sixth energy levels for an electron in potential well 0.1nm wide.

Solution:

$$\text{Energy (E)} = \frac{n^2 \hbar^2}{8mL^2}$$

4. For an electron confined to an infinite potential well of width 0.1nm, determine the uncertainty in momentum and kinetic energy.

Solution:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

5. If De-Broglie wavelength of an electron is 980nm. What is its velocity?

Solution:

$$\lambda = \frac{h}{p}$$

6. Calculate the fermi energy at 0K for copper given its density is 8.96g/cm³ and atomic mass 63.5 g/mol.

Solution:

$$n = \frac{\rho N_A}{m}$$

$$E_F = \frac{h^2}{2m} \left(\frac{3n}{8\pi} \right)^{\frac{2}{3}}$$

7. Determine the ground state energy of an electron which is confined in a well of width 8.5nm. How this electron be put to the fourth energy level?

Solution:

$$E = \frac{n^2 \hbar^2}{8mL^2}$$

8. If longest wavelength required photoelectric effect for a certain metal is 2250Å.

i. Find work function of the metal

$$\phi = h \frac{c}{\lambda}$$

ii. The velocity of electrons ejected from the metal surface if the same metal is radiated with $\lambda = 1800\text{Å}$.

$$E = h \frac{c}{\lambda}$$

$$h \frac{c}{\lambda} - \phi = \frac{mv^2}{2}$$

iii. Kinetic energy of ejected electrons.

$$\text{Kinetic energy} = h \frac{c}{\lambda} - \phi$$

9. An electron is confined to an infinite potential well of size 0.1nm. Calculate the ground energy of the electron and radian frequency.

Solution:

$$E = \frac{n^2 \hbar^2}{8mL^2}$$

$$E = \hbar \omega$$

10. Calculate the wavelength of a proton travelling at 1900 ms⁻¹.

Solution:

$$\lambda = \frac{h}{p}$$

Chapter 2 and 3

Important formulae

$$KE = \frac{1}{2}mv^2 = \frac{3kT}{2}$$

$$v_D = \mu_D E$$

$$\mu_D = \frac{e\tau}{m}$$

$$J = eN_e \mu_D E$$

$$\sigma = eN_e \mu_D$$

$$R = \frac{\rho l}{A}$$

1. The resistivity of copper at 20 degree centigrade is $0.69 \times 10^{-8} \Omega m$ and density of free electron is $8.5 \times 10^{28} m^{-3}$. Calculate the mean free time of electron in copper lattice. Assume $m^* = 1.01m$, where $m = 9.1 \times 10^{-31} kg$.

Solution:

$$\rho = \frac{1}{\sigma}$$

$$\sigma = eN_e\mu_D$$

$$\mu_D = \frac{e\tau}{m}$$

2. If electrical conductivity of potassium is $1.39 \times 10^5 Sm/cm$ calculate the drift mobility of electron at room temperature. Molar mass and density of potassium are 39.95 and 0.91 gm/cc.

Solution:

$$N_e = \frac{\rho N_A}{M_{at}}$$

$$\sigma = eN_e\mu_D$$

3. Calculate the mean free path and the mean free time between collisions of an electron having drift mobility $43 cm^2 v^{-1} s^{-1}$ and mean speed $1.2 \times 10^6 ms^{-1}$.

Solution:

$$\mu_D = \frac{e\tau}{m}$$

$$v_D = \frac{d}{\tau}$$

4. Calculate the drift mobility and the mean scattering time of conduction electrons in copper at room temperature, given that the conductivity of copper is $5.9 \times 10^5 \Omega^{-1} cm^{-1}$. The density of copper is $8.96 gm/cm^3$ and its atomic mass is $63.5 gm/mol$. (assume one free electron per atom)

Solution:

$$N_e = \frac{\rho N_A}{M_{at}}$$

$$\sigma = eN_e\mu_D$$

5. A uniform silver wire has resistivity of $5 \times 10^{-8} \Omega m$ at room temperature. For the electrical field the wire of $1 V/cm$, compute average drift velocity of electron assuming there are 5.8×10^{28} conduction electron per m^3 . Also calculate mobility and relaxation time of electron.

Solution:

$$\sigma = eN_e\mu_D$$

$$v_D = \mu_D E$$

$$\mu_D = \frac{e\tau}{m}$$

6. Molybdenum has the BCC crystal structure, a density of 10.22 g cm^{-3} , and an atomic mass of 95.94 g mol^{-1} . What is atomic concentration, lattice parameter and atomic radius of molybdenum?

Solution:

BCC crystal structure contains 2 atoms per unit cell.

$$N_e = \frac{\rho N_A}{M_{at}}$$

$$\text{lattice parameter } (a) = \left[\frac{NM}{N_A \rho} \right]^{\frac{1}{3}}$$

$$a = \frac{4R}{\sqrt{3}}$$

Chapter 4 and 5

Important Formulae

$$C = \frac{\epsilon_r \epsilon_0 A}{d}$$

$$P = \epsilon_0 \chi_e E$$

$$\frac{(\epsilon_r - 1)}{(\epsilon_r + 2)} = \frac{N \alpha_e}{3 \epsilon_0}$$

$$\epsilon_r = \frac{C}{T - T_0}$$

$$\tan \delta = \frac{\epsilon_r''}{\epsilon_r'}$$

$$\text{Magnetization}(M) = \text{susceptibility}(\chi) \times \text{field strength}(H)$$

$$\text{Flux density}(B) = \text{permeability}(\mu) \times \text{field strength}(H)$$

$$\text{Magnetic field inside the solenoid } (B_0) = \mu_0 n I = \mu_0 H$$

$$B = \mu_0 \mu_r H (\text{Magnetizing field})$$

$$\text{flux density } (B) = \frac{\Phi}{A}$$

$$\text{Magnetic field } (M) = \chi_m H$$

1. A parallel plate capacitor of area 20 cm^2 and separation of 1 mm. The space between plates is filled with Al_2O_3 when potential difference of 10V is applied. Calculate field strength and dipole moment induced in oxide layer. Assume relative permittivity of $\text{Al}_2\text{O}_3 = 8$.

Solution:

$$E = \frac{V}{d}$$

$$C = \frac{\epsilon_r \epsilon_0 A}{d}$$

$$C = \frac{Q}{V}$$

$$p = Qd$$

2. The optical index of reflection and the dielectric constant for glass are 1.45 and 6.5 respectively. Calculate the percentage of Ionic polarizability.

Solution:

$$\frac{(\epsilon_r - 1)}{(\epsilon_r + 2)} = \frac{N(\alpha_e + \alpha_i)}{3\epsilon_0}$$

$$\frac{(n^2 - 1)}{(n^2 + 2)} = \frac{N\alpha_e}{3\epsilon_0}$$

3. A parallel plate capacitor has an area of 10 cm^2 and a separation of 0.1mm. The space between the plates is filled with polyethylene. An alternative voltage with amplitude of 2 volts is applied at a frequency of 1MHz. At this frequency the real part of the relative permittivity is 2.25 and the loss tangent is 4×10^{-4} . Calculate the energy dissipated per second.

Solution:

$$E = \frac{V}{d}$$

$$P = \omega E^2 \epsilon_0 \epsilon_r' \tan \delta v$$

4. Calculate the magnetization and flux density in a diamagnetic sample having susceptibility -0.3×10^{-5} and magnetic field strength 1000A/m.

Solution:

$$\text{Magnetization}(M) = \text{susceptibility} \times \text{field strength}(H)$$

$$\text{Flux density}(B) = \text{permeability} \times \text{field strength}(H)$$

5. The electronic polarizability of Ar atom $1.7 \times 10^{-4} \text{ Fm}^2$. Calculate the static dielectric constants of solid of Ar given its density is 1.8 gcm^{-3} , $M_{at} = 39.5 \text{ gmmol}^{-1}$ and $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$.

Solution:

$$\frac{(\epsilon_r - 1)}{(\epsilon_r + 2)} = \frac{N\alpha_e}{3\epsilon_0}$$

6. Find the percentage increase in magnetic field B, when the space within a current carrying coil is fixed with aluminum. The susceptibility of aluminum is 2.132×10^{-5} .

Solution:

$$\frac{B - B_0}{B_0} \times 100$$

$$B = \mu_0 \mu_r H$$

$$H = nI$$

7. A electrolyte condenser consisting of a oxidizing aluminum sheet with effective surface area 400cm^2 has capacitance of $8\mu\text{F}$. Relative dielectric constant of Al_2O_3 is 8. The potential difference of 10 V is applied between aluminum and electrode. The field strength and dipole moment induced in oxide layer?

Solution:

$$E = \frac{V}{d}$$

$$C = \frac{\epsilon_r \epsilon_0 A}{d}$$

$$C = \frac{Q}{V}$$

$$p = Qd$$

8. The magnetic fields strength in a piece of Fe_2O_3 is 10^6 Am^{-1} . Given that the susceptibility at room temperature is 1.4×10^{-3} , find the flux density and magnetization in the material.

Solution:

$$B = \mu_0 \mu_r H$$

$$M = \frac{B}{\mu_0}$$

9. Calculate the permeability and susceptibility of an iron bar of cross-sectional area 0.2cm^2 when a magnetic field strength of 1200A/m produces magnetic flux of 24 micro weber.

Solution:

$$\text{flux density } (B) = \frac{\Phi}{A}$$

$$B = \mu H$$

$$\chi_m = \mu_r - 1 = \frac{\mu}{\mu_0} - 1$$

10. Calculate the field intensity of a magnetic field and the intensity of magnetization if 0.2A current is passed through a winding of 20 turns/cm over an iron anchor ring having magnetic field 1.26T.

Solution:

$$\text{Field intensity } (H) = nI$$

$$\text{Intensity of magnetization } (M) = \frac{B}{\mu_0}$$

Chapter 6

1. Calculate built in potential across the junction formed by joining silicon doped with 10^{17} donor atom per cm^3 with silicon doped with 10^{16} acceptor atom per cm^3 at room temperature. Take the intrinsic concentration as $1.45 \times 10^{10} \text{cm}^{-3}$.

Solution:

$$V_0 = \frac{kT}{e} \ln\left[\frac{n_{n_0}}{n_{p_0}}\right]$$

$$n_{n_0} = N_d$$

$$n_{p_0} = \frac{n_i^2}{N_A}$$

2. A heavily doped p-side with acceptor concentration of 10^{18} cm^{-3} is connected to n-side with donor concentration of 10^{16} cm^{-3} . Calculate the built in potential, depletion width in n-side and p-side and overall depletion width. Assume, $T = 300\text{K}$ and $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$.

Solution:

$$V_0 = \frac{eN_a D^2 N_d}{2\epsilon(N_a + N_d)}$$

$$V_0 = \frac{kT}{e} \ln\left[\frac{n_{n_0}}{n_{p_0}}\right]$$

$$n_{n_0} = N_d$$

$$n_{p_0} = \frac{n_i^2}{N_A}$$

3. N-type semiconductor doped with 10^{16} cm^{-3} phosphorous atoms has been doped with 10^{17} cm^{-3} boron atoms. Calculate the electron and hole concentration in the semiconductors.

Solution:

$$p = N_a - N_d$$

$$n = \frac{n_i^2}{p}$$

4. Calculate the diffusion coefficient of electrons at 300K in n-type silicon semiconductor doped with 10^{15} arsenic atoms per cm^3 . (μ for given doping $1300 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$)

Solution:

$$\frac{D_e}{\mu_e} = \frac{kT}{e}$$

5. Suppose a P-N junction is created on silicon whose relative permittivity (ϵ_r)=2.3. If the donor level an N-side 10^{17} cm^{-3} and acceptor level on P-side is 10^{16} cm^{-3} . Calculate built in potential (V_0), and junction width. [$n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$]

Solution:

$$V_0 = \frac{eN_a D^2 N_d}{2\epsilon(N_a + N_d)}$$

$$V_0 = \frac{kT}{e} \ln\left[\frac{n_{n_0}}{n_{p_0}}\right]$$

$$n_{n_0} = N_d$$

$$n_{p_0} = \frac{n_i^2}{N_A}$$

6. A Si sample has been doped with 10^{17} arsenic atoms cm^{-3} . Calculate the conductivity of sample at $27^\circ C$ and $127^\circ C$. Given $\mu_e = 800 cm^2 V^{-1} s^{-1}$ at $27^\circ C$, $\mu_e = 420 cm^2 V^{-1} s^{-1}$ at $127^\circ C$ and $n_i = 1.45 \times 10^{10} cm^{-3}$.

Solution:

$$\sigma = en\mu_e$$